

Question-1

The Problem Statement

Title: KSPL Franchise Management System

Scenario: You are tasked with designing the backend logic for a cricket tournament registration system. You need to create a **Franchise** class that models a participating team, and a **FranchiseTester** class to run the execution logic.

Requirements for the **Franchise class:**

- It must store the franchise's name, maximum squad size, current budget, and the records of the players (names and jersey numbers) currently signed.
- **initializeFranchise(String name, int squadLimit, double budget)**: Sets up the team's initial information and arrays.
- **signPlayer(String playerName, int jerseyNumber, double cost)**: Attempts to add a player to the team. You must check the following conditions in this exact order:
 1. If the team has reached its **squadLimit**, print: **Cannot sign [playerName]: Squad is full!**
 2. If the team has space but another player already has the requested jersey, print: **Cannot sign [playerName]: Jersey #[jerseyNumber] is already taken!**
 3. If the jersey is free but the cost is greater than the team's current budget, print: **Cannot sign [playerName]: Insufficient budget!**
 4. If all checks pass, add the player, deduct the cost from the budget, and print: **[playerName] has joined the squad!**
- **increaseBudget(double extraFunds)**: Adds the specified amount to the team's current budget and prints: **[Franchise Name] budget increased by \$[extraFunds]**
- **teamDetails()**: Prints the team's name, remaining budget, squad status, and a numbered list of players with their jersey numbers. If no players are signed, print: **No players signed yet.**

Requirements for the **FranchiseTester class:** Write the **Driver Code** exactly as provided below to test your **Franchise** class. Your output must perfectly match the

Expected Output:

```
public class FranchiseTester {
    public static void main(String[] args){
        Franchise f1 = new Franchise();
        Franchise f2 = new Franchise();

        f1.initializeFranchise("Deca
Limited", 3, 500.0);
        f2.initializeFranchise("Strikers",
2, 1000.0);

        System.out.println("1 =====");
        f1.teamDetails();

        System.out.println("2 =====");
        f2.signPlayer("Alen", 10, 400.0);
        f2.signPlayer("Ahnaf", 17, 450.0);
        f2.teamDetails();

        System.out.println("3 =====");
        f2.signPlayer("Sara", 5, 100.0);

        System.out.println("4 =====");
        f1.signPlayer("ROUDRO", 7, 200.0);
        f1.signPlayer("Afrida", 7, 150.0);
        f1.signPlayer("Afrida", 11, 400.0);

        System.out.println("5 =====");
        f1.increaseBudget(200.0);
        f1.signPlayer("Afrida", 11, 400.0);

        System.out.println("6 =====");
        f1.teamDetails();
    }
}
```

```
1 =====
Team: Deca Limited
Budget: $500.0
Squad Status: 0/3 players
No players signed yet.
2 =====
Alen has joined the squad!
Ahnaf has joined the squad!
Team: Strikers
Budget: $150.0
Squad Status: 2/2 players
Roster:
1. Alen (Jersey: 10)
2. Ahnaf (Jersey: 17)
3 =====
Cannot sign Sara: Squad is full!
4 =====
ROUDRO has joined the squad!
Cannot sign Afrida: Jersey #7 is
already taken!
Cannot sign Afrida: Insufficient
budget!
5 =====
Deca Limited budget increased by $200.0
Afrida has joined the squad!
6 =====
Team: Deca Limited
Budget: $100.0
Squad Status: 2/3 players
Roster:
1. ROUDRO (Jersey: 7)
2. Afrida (Jersey: 11)
```